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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia							A																																																		
B								B																																																		
C	Project Name : MV Switchgear for MNA Serang Plant							C																																																		
	Switchgear Name : MVD-32 20kV Tank Farm WICA																																																									
D	Panel Type : UNISEC							D																																																		
	Customer Name : PT Multimas Nabati Asahan																																																									
E	Contract No. : BE318042							E																																																		
	TITLE : SLD & GA Drawing																																																									
F								F																																																		
<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td colspan="2">PROJECT TITLE</td><td rowspan="2">ABB</td><td rowspan="2">EPMV</td><td rowspan="4">COVER SHEET SLD & GA Drawing MVD-32 20kV Tank Farm WICA</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td colspan="2">MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td colspan="2"></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>																Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	COVER SHEET SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant		Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW			Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.		Cont. 16
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A
B	SWITCHGEAR VERSION			= UNISEC	COLOUR OF FRONT DOORS = RAL 7035				B
	RATED VOLTAGE (Un)			= 24 kV	SIDEWALLS PAINTED = YES				
	OPERATING VOLTAGE			= 20 kV	SIDES = L+R				
	RATED FREQUENCY (fr)			= 50 Hz					
	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)			= 125 kV					
C	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)			= 50 kV					C
	RATED CURRENT OF MAIN BUSBARS (Ir)			= 630 A					
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)			= 20 kA					
	RATED PEAK WITHSTAND CURRENT (Ip)			= 50 kA					
	RATED DURATION OF SHORT CIRCUIT (tk)			= 1 s					
D	INTERNAL ARC CLASSIFICATION (IAC)			= AFLR / 16kAx1s					D
	AMBIENT CONDITION			= NORMAL					
	AMBIENT AIR TEMPERATURE			= -5°C...+40°C					
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)			= IP3X					
	DEGREE OF PROTECTION WITH OPEN DOORS			= IP2X					
E	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)			= 230VAC					E
	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)			= 230VAC					
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)			= 230VAC					
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= 450/750V					
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= PVC					
F	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS			= 2.5 mm²/BLACK					F
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS			= 1.5 mm²/BLACK					
	CROSS-SECTION OF CONDUCTORS FOR MOTORS			= 2.5 mm²/BLACK					
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)			= 1.5 mm²/BLACK					
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS			= 1.5 mm²/BLACK					
CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE			= 2.5 mm²/BLACK						
COMMUNICATION CABLE			= N/A						
1 Issued for Approval 18.12.2018 YR Date Prep. 15.11.2018 PROJECT TITLE MV Switchgear for MNA Serang Plant Based on: BE318042 ABB EPMV PT. ABB Sakti Industri GENERAL CHARACTERISTICS SLD & GA Drawing MVD-32 20kV Tank Farm WICA Scale Customer Drawing Nbr. Customer Doc. Nbr. = H00 PT Multimas Nabati Asahan Sh. 3 Cont. 16									
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FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW

-XDA:10

91011

-XDA:10

(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)

-BGB5	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED
-BGB6	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY
-BGB7	POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR
-BGB8...-BGB10	POSITION SWITCHES OF CIRCUIT-BREAKER
-BGB11	POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING
-BGB12	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION
-BGB13	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION
-BGD	POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.
-BGD1	POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR
-BGD2	POSITION SWITCH OF CABLE COMPARTMENT DOOR
-BGE1	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION
-BGE2	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION
-BGE3	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGE4	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR
-BGE5	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR
-BGE7	POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION
-BGE8	POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS
-BGE11	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION
-BGE12	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION
-BGE13	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGE14, -BGE15	POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR
-BGF	POSITION SWITCHES OF MEDIUM VOLTAGE FUSES
-BGI1	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION
-BGI2	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION
-BGI3	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGI4	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR
-BGI5	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR
-BGI6	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION
-BGI7	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION
-BGI8	POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS
-BGI9	POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR
-BGI11	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION
-BGI12	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION
-BGI13	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGI14, -BGI15	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR
-BGI21	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION
-BGI22	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION
-BGI23	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)
-BGI24, -BGI25	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR

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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION		-CA		CAPACITORS	
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION		-CA1		CAPACITOR FOR FRONT VENTILATOR	
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		-CA2		CAPACITOR FOR REAR VENTILATOR	
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION		-EA1		LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.	
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION		-EA2		LIGHTING LAMP LOCATED IN CABLE COMPARTMENT	
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION		-EA3		LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE	
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION		-EA4		LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT	
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION		-EB1		HEATER LOCATED IN CABLE COMPARTMENT	
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION		-EB2		HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT	
	-BGK		POSITION SWITCH OF KEY LOCK		-EB3		HEATER LOCATED IN MOTOR CABINET	
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1		-EB4		HEATER FOR VENTILATION TEST	
	-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS		-EB5		HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.	
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION		-EB7		HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE	
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION		-EB8		HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE	
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION		-FA1		SURGE ARRESTER LOCATED ON PHASE L1	
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION		-FA2		SURGE ARRESTER LOCATED ON PHASE L2	
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION		-FA3		SURGE ARRESTER LOCATED ON PHASE L3	
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION		-FCD		FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS	
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION		-FCF1		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1	
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION		-FCF2		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2	
C	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION		-FCF3		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3	
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION		-FCM1		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER	
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION		-FCM2...-FCM10		MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS	
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR		-FCM11		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1	
	-BGV2		POSITION SWITCH OF REAR VENTILATOR		-FCM12		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2	
	-BM		HYGROSTAT		-FCM13		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3	
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-FCT		THERMAL OVERLOAD RELEASES	
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GA		GENERATORS	
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ1		FRONT VENTILATOR	
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ2		REAR VENTILATOR	
D	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER		-GQ3		VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK	
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2		-KFA1		AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE	
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3		-KFA2		AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE	
	-BR		FLAME DETECTORS, SMOKE DETECTORS		-KFA3....-KFA9		AUXILIARY RELAYS OR CONTACTORS	
	-BT		THERMOSTAT		-KFC		CLOSING RELAYS OR CONTACTORS	
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS		-KFI		INTEGRATED CIRCUITS	
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS		-KFL		LOCKOUT RELAY	
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM		-KFM		MICROPROCESSORS	
	-BU		BUCHHOLZ RELAY		-KFN		ANTIPUMPING RELAYS	
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING		-KFO		OPENING RELAYS OR CONTACTORS	
E	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING		-KFP		PROGRAMMABLE LOGIC CONTROLLERS (PLC)	
	-BZ,-BZ4		MULTIFUNCTION SENSORS		-KFR		RECLOSING RELAYS	
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1		-KFS		SYNCHRONIZING DEVICES	
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS	
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS	
					-KFZ		CONTROLLERS	
					-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER	
					-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD	
					-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1	
					-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2	
F					-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS	

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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

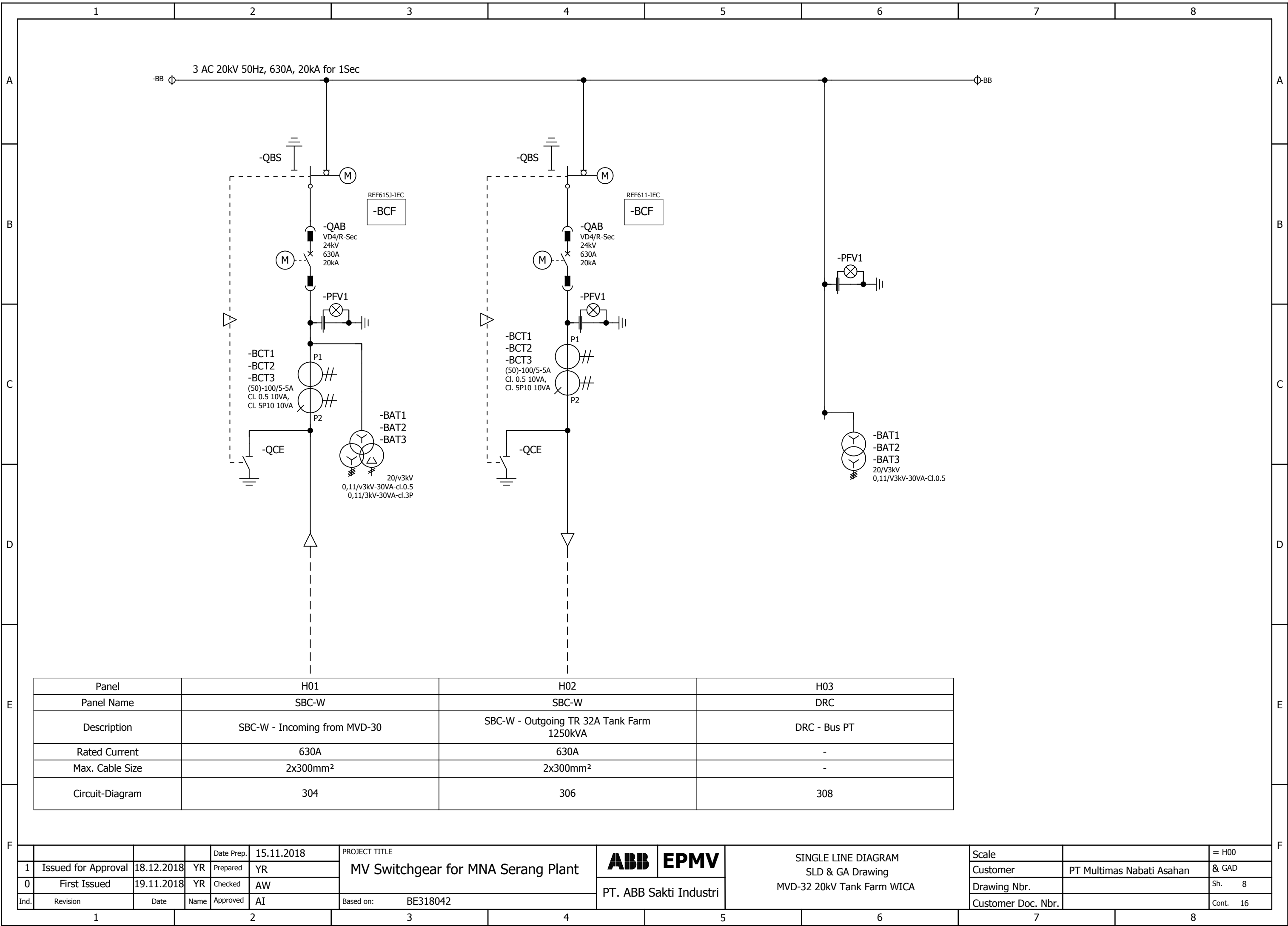
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant				SLD & GA Drawing		Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri		MVD-32 20kV Tank Farm WICA		Drawing Nbr.		Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042						Customer Doc. Nbr.		Cont. 16

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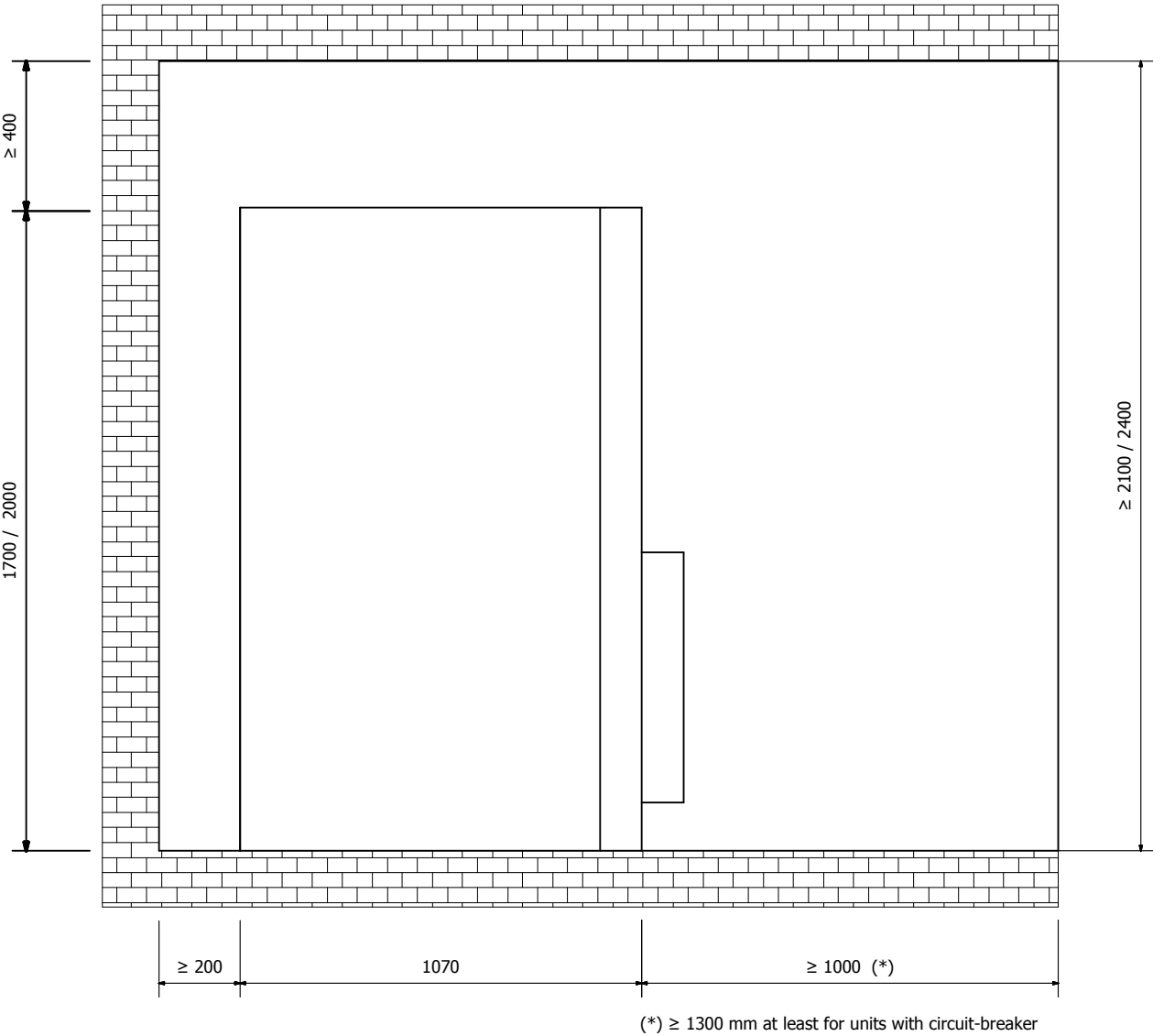
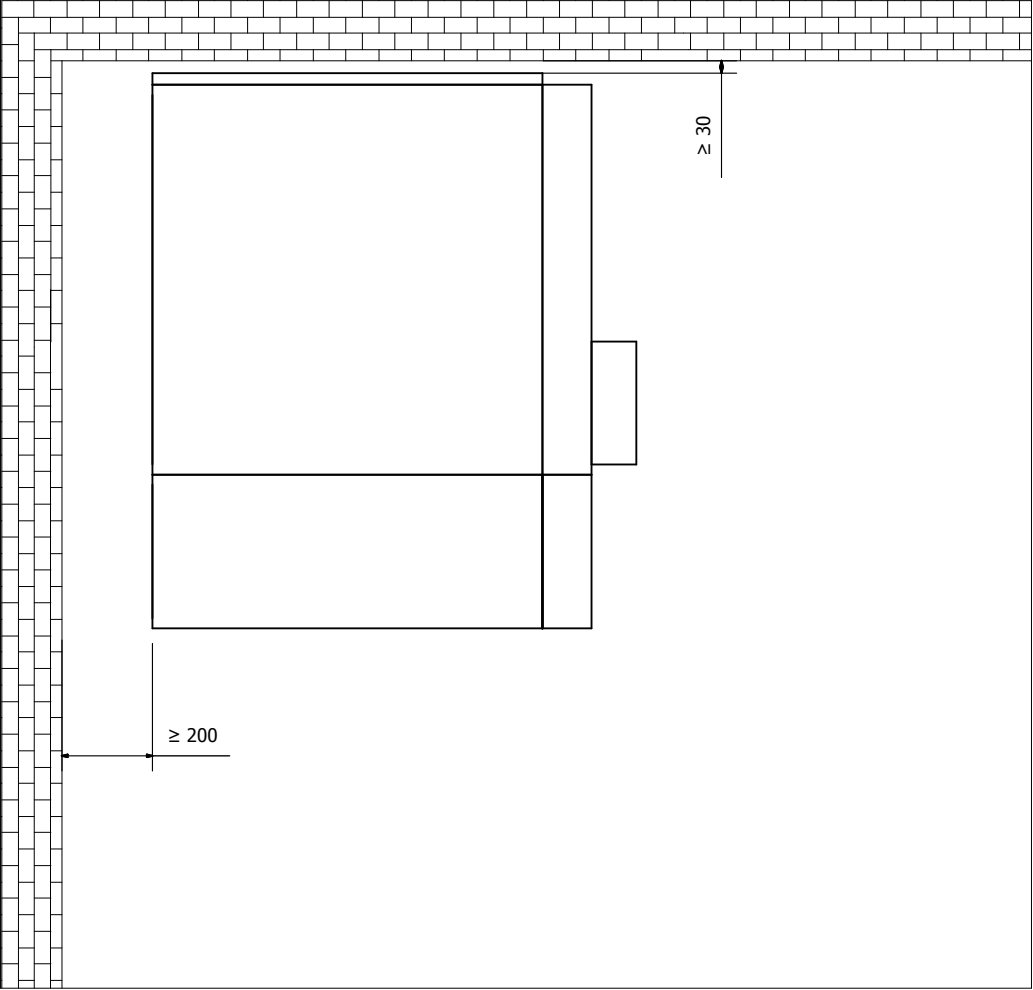
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RULES FOR INSTALLATION IN THE ROOM

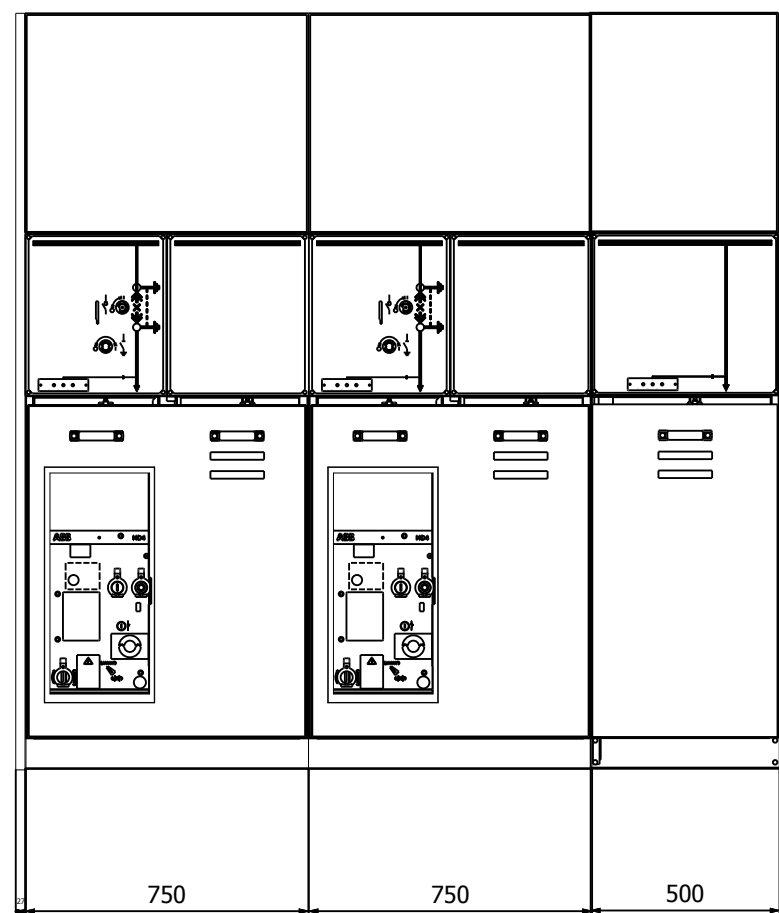
MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INSTALLATION RULES SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 9	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont. 16	

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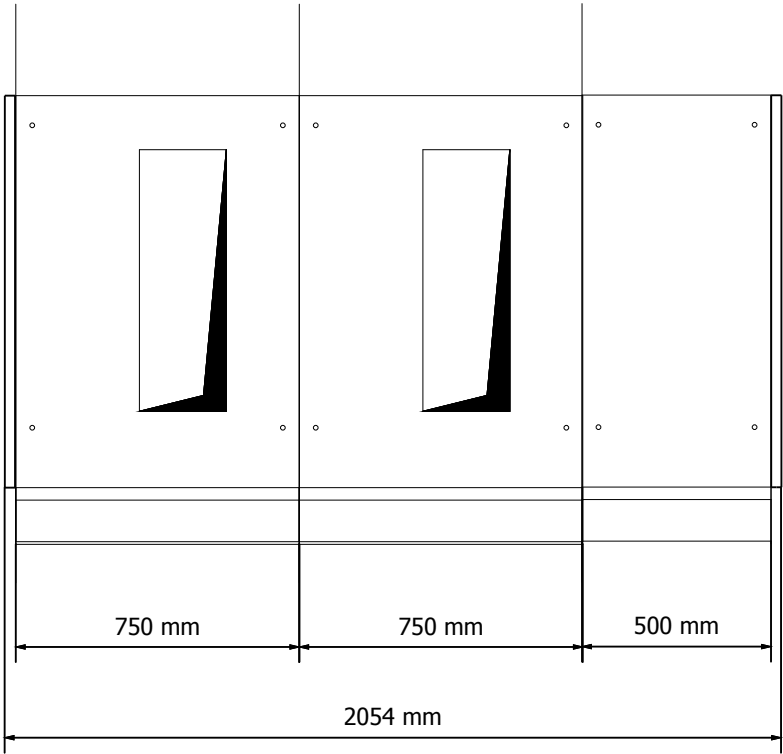


Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 10	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont. 16	

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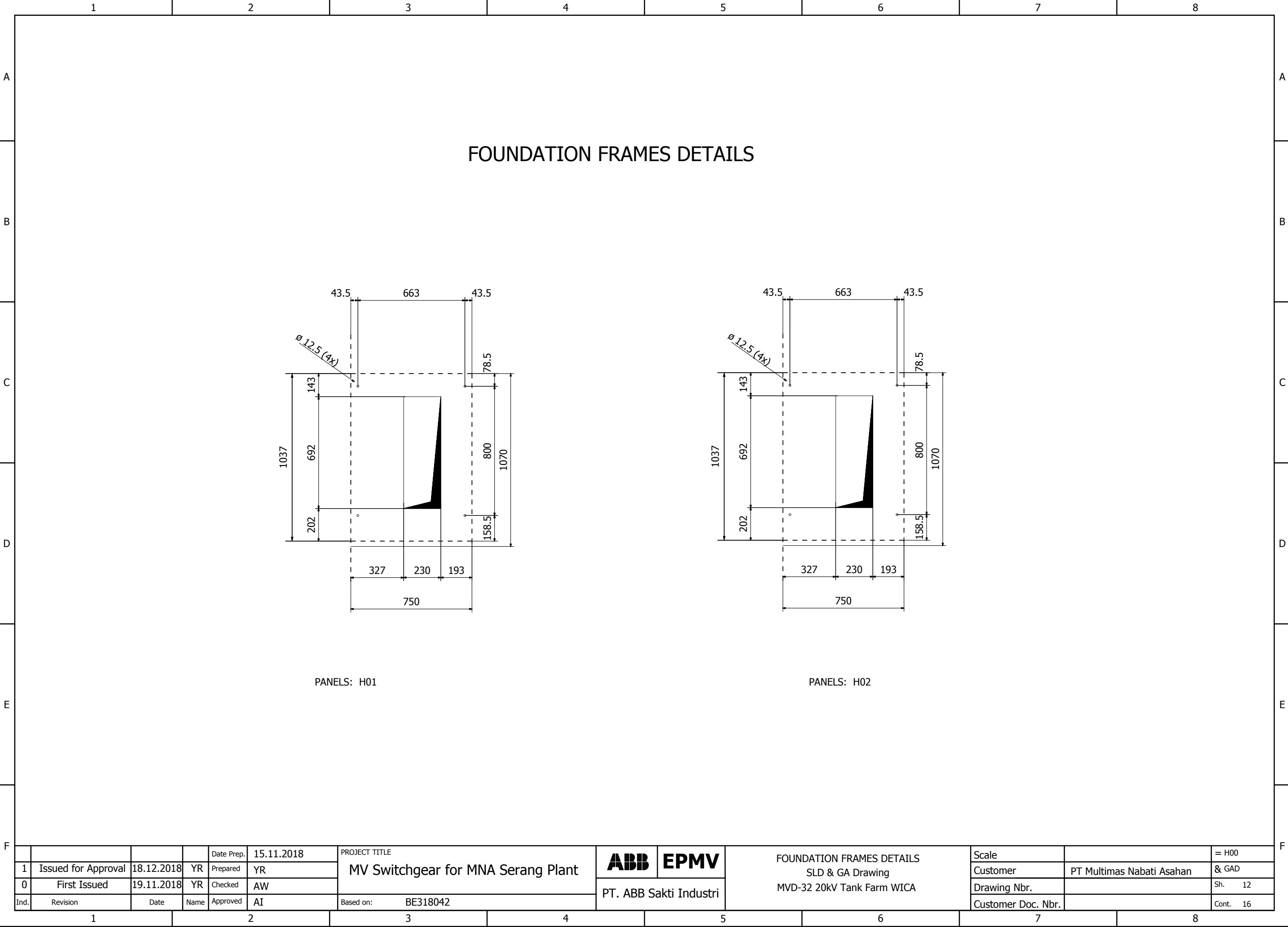
FOUNDATION FRAMES



Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 11		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 12		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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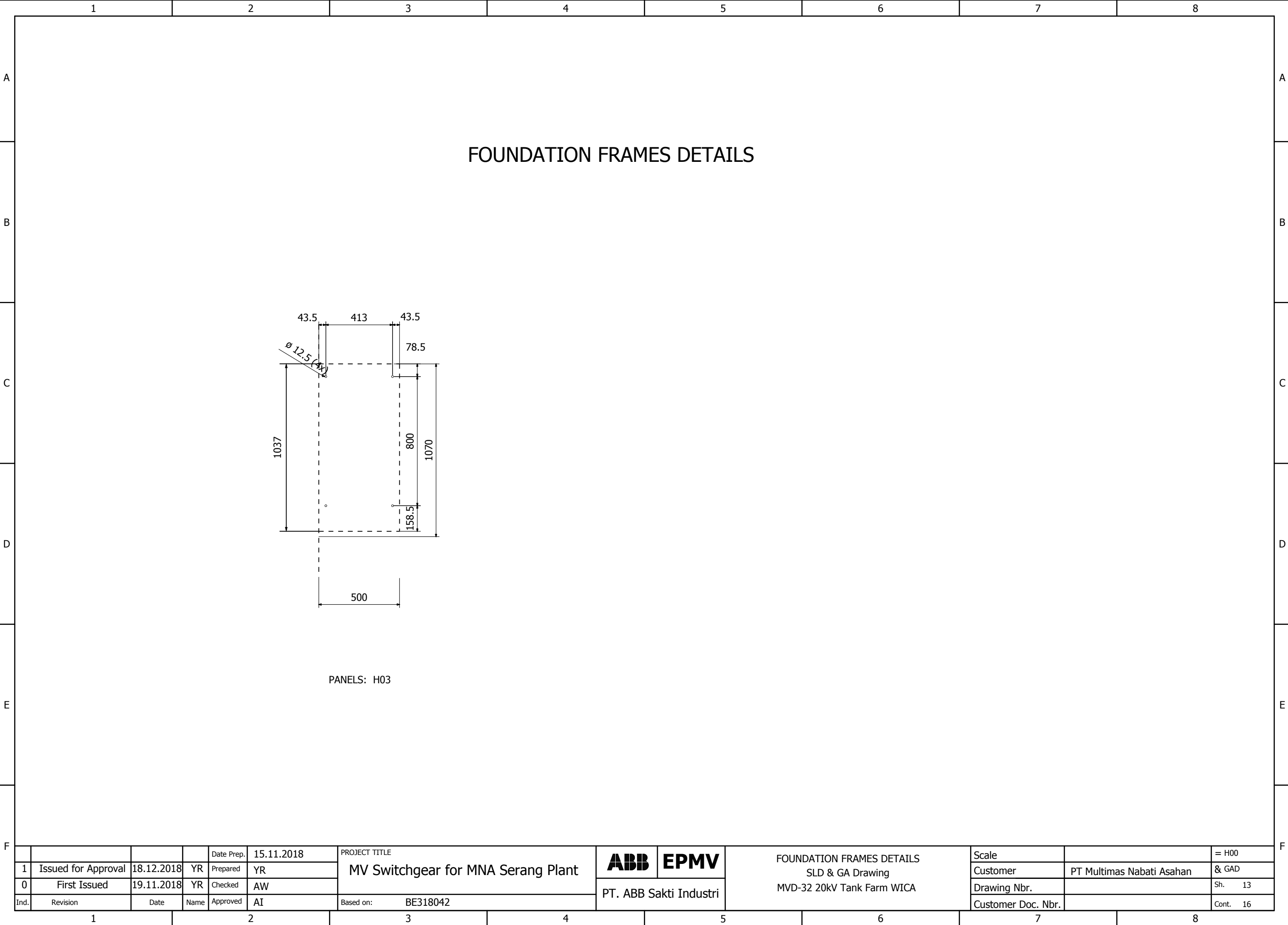
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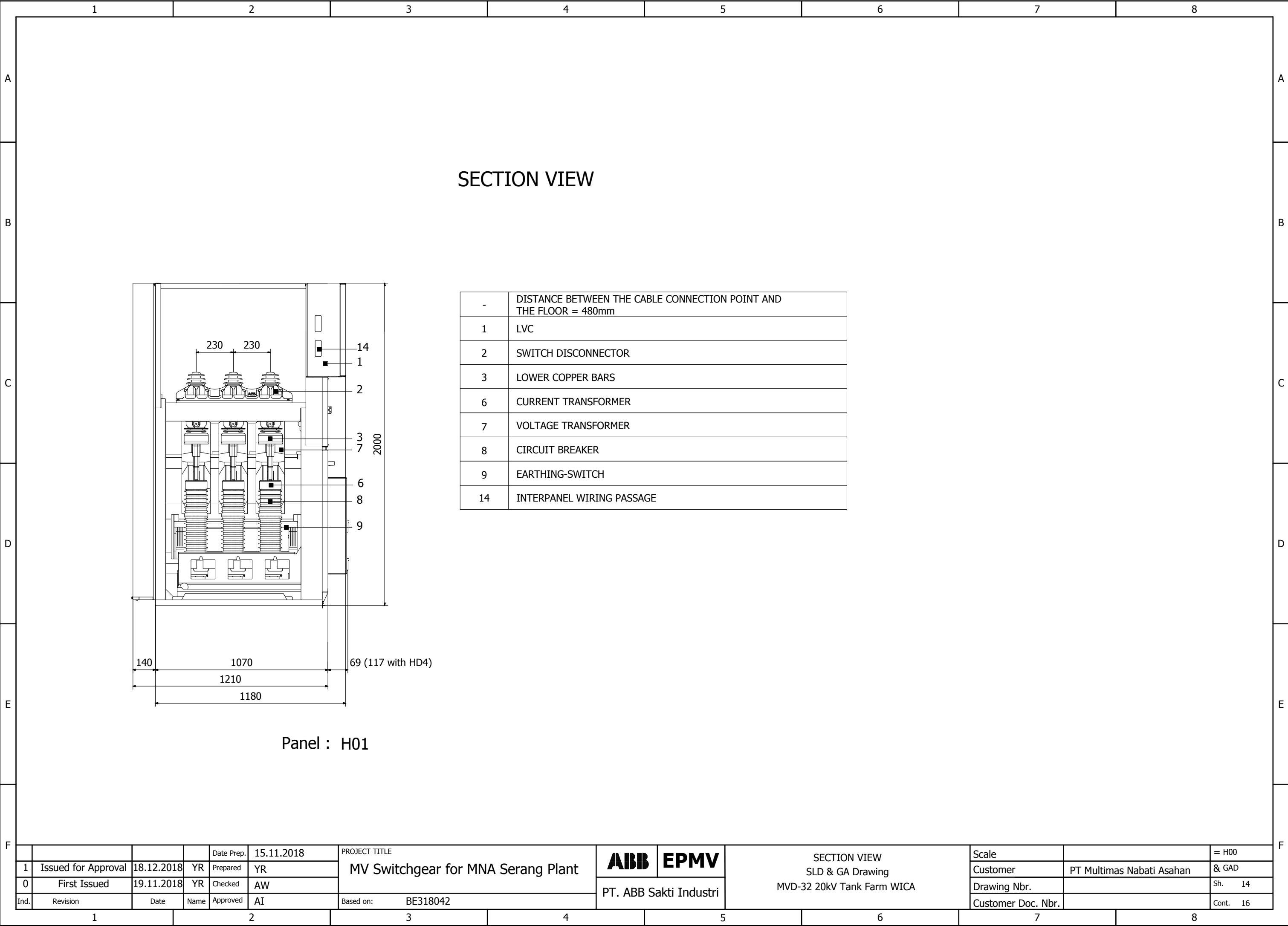
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-32 20kV Tank Farm WICA	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 14		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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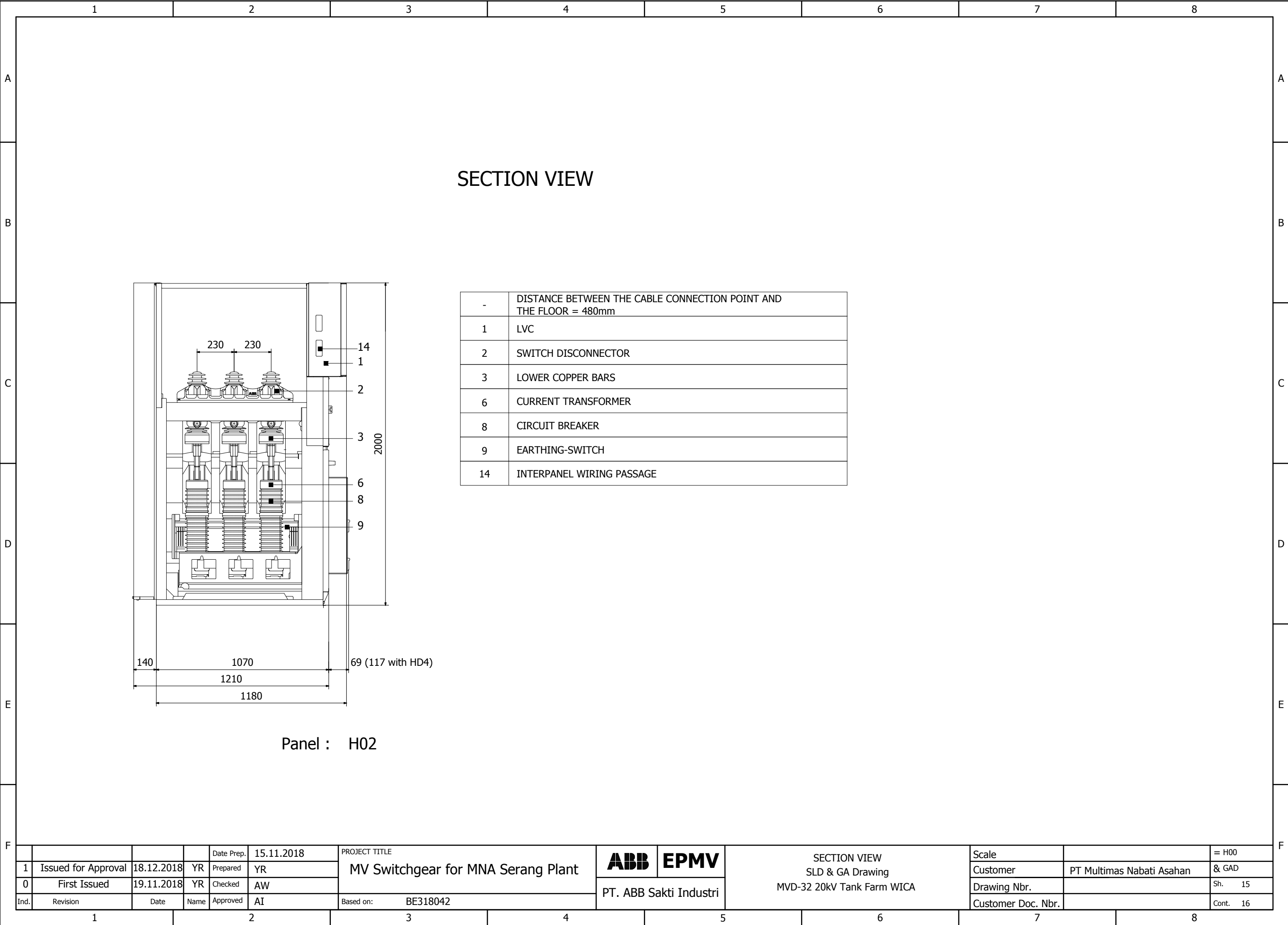
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